GENERAL NOTES

1. THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS AT SITE AND SHALL NOTIFY THE ENGINEER ON RECORD OF ANY DISCREPANCY BETWEEN THE ACTUAL CONDITION AND THE INFORMATION SHOWN ON THE DRAWINGS. THESE SHALL INCLUDE BUT NOT LIMITED TO THE LOCATIONS AND DIMENSIONS OF GROOVES, CHASES, SLEEVES, CURBS, OPENINGS, EMBEDDED OR ATTACHED ITEMS, ETC. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND PLUMBING DRAWINGS.
2. IN THE INTERPRETATION OF THESE DRAWINGS, INDICATED DIMENSIONS SHALL TAKE PRECEDENCE OVER SCALED MEASUREMENTS OF PLANS, SECTIONS AND DETAILS. NOTES AND DETAILS ON DRAWINGS SHALL TAKE PRECEDENCE OVER THE GENERAL NOTES AND THE TYPICAL DETAILS.
3. THE STRUCTURAL DRAWINGS HEREIN REPRESENT THE FINISHED STRUCTURE. THEY DO NOT INDICATE THE METHOD OF CONSTRUCTION UNLESS SO STATED. THE CONTRACTOR SHALL THEREFORE PROVIDE ALL THE NECESSARY MEASURES IN ORDER TO PROTECT THE STRUCTURE, THE ADJACENT PROPERTIES, WORKMEN AND ALL OTHER PERSONS AT SITE DURING ALL PHASES OF THE CONSTRUCTION. THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE ENGINEER ON RECORD OF ANY CONDITION WHICH IN HIS OPINION MIGHT ENDANGER THE STABILITY OF THE STRUCTURE OR MIGHT CAUSE DISTRESS TO THE STRUCTURE.
4. CONSTRUCTION MATERIALS SHALL NOT BE STORED ON POURED SLABS. IT IS THE RESPONSIBILITY OF THE GENERAL CONTRACTOR TO ENSURE THAT THE SUB-CONTRACTOR/S (IF THERE ARE ANY) ARE INFORMED OF THIS SO AS NOT TO VIOLATE THIS REQUIREMENT.
5. THE CONTRACTOR SHALL PROVIDE ALL TEMPORARY GUYING AND/OR BRACING REQUIRED TO ERECT AND HOLD THE STRUCTURE IN PROPER ALIGNMENT UNTIL ALL STRUCTURAL WORKS AND CONNECTIONS HAVE BEEN COMPLETED. THE INVESTIGATION, DESIGN, SAFETY, ADEQUACY AND INSPECTION OF ERECTION BRACING, SHORING, TEMPORARY SUPPORTS, ETC. ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR.
6. THE CONTRACTOR SHALL TAKE ALL THE NECESSARY STEPS TO ENSURE THE PROPER ALIGNMENT OF THE STRUCTURE AFTER THE INSTALLATION OF ALL STRUCTURAL AND FINISHED MATERIALS.
7. UNLESS OTHERWISE SHOWN, ALL DIMENSIONS ARE IN METERS.
8. ALL COORDINATES AND ELEVATIONS ARE IN METERS.
9. ALL CONCRETE WORK SHALL BE DONE IN ACCORDANCE WITH THE LATEST ACI BUILDING CODE FOR REINFORCED CONCRETE AND ALL STRUCTURAL STEEL WORK SHALL BE IN ACCORDANCE WITH THE LATEST AISC SPECIFICATIONS IN SO FAR AS THEY DO NOT CONFLICT WITH THE LOCAL BUILDING REQUIREMENTS.

CONCRETE

1. UNLESS OTHERWISE INDICATED IN PLANS OR NOTED IN THE SPECIFICATIONS THE MINIMUM 28-DAY CYLINDER COMPRESSIVE STRENGTH OF CONCRETE SHALL BE AS FOLLOWS:

1. FOOTINGS, FOOTING TIE BEAMS AND SLAB ON GRADE	f'c = 21.00 MPa (3000 psi)
2. PEDESTAL, COLUMNS AND R.C. WALLS	f'c = 21.00 MPa (3000 psi)
3. R.C. FLOOR SLABS, BEAMS AND GIRDERS	f'c = 21.00 MPa (3000 psi)
4. LEAN CONCRETE	f'c = 14.00 MPa (2000 psi)

2. AGGREGATE SIZE SHALL BE AS FOLLOWS:

1. COLUMNS, GIRDERS, BEAMS, SLABS AND FOOTINGS	20 mm MAXIMUM
2. FOOTINGS AND SLAB ON GRADE	40 mm MAXIMUM

3. MINIMUM CONCRETE COVER FOR REINFORCING BARS SHALL NOT BE LESS THAN THE FOLLOWING:

1. UNFORMED CONCRETE DEPOSITED AGAINST THE GROUND	75 mm
2. FORMED CONCRETE EXPOSED TO GROUND OR WEATHER WHEN BARS ARE GREATER THAN ø16 MM	50 mm
3. FORMED CONCRETE EXPOSED TO GROUND OR WEATHER WHEN BARS ARE ø16 MM OR SMALLER	40 mm
4. BEAMS OR COLUMNS NOT EXPOSED TO GROUND OR WEATHER	40 mm
5. SUSPENDED SLABS AND WALLS NOT EXPOSED TO GROUND OR WEATHER	20 mm
6. CONCRETE EXPOSED TO SEWAGE	75 mm

4. ALL REINFORCING STEEL, DOWELS, ANCHOR BOLTS AND OTHER INSERTS SHALL BE SECURED IN POSITION PRIOR TO POURING OF CONCRETE.

REINFORCING BARS

1. ALL REINFORCING BARS SHALL BE FABRICATED AND INSTALLED IN CONFORMANCE WITH THE MANUAL OF STANDARD PRACTICE FOR DETAILING REINFORCED CONCRETE STRUCTURES, ACI 315 LATEST EDITION.
2. ALL REINFORCING STEEL BARS SHALL BE NEW BILLET, HOT ROLLED, DEFORMED BARS CONFORMING TO THE SPECIFICATIONS OF PNS 49:2002 WHOSE MINIMUM YIELD STRENGTH SHALL BE 275 MPA (GRADE 40) FOR RSB 12mmø AND 414 MPA (GRADE 60) FOR RSB 16mmø OR GREATER.
3. ALL REINFORCING BARS SHALL BE ACCURATELY AND SECURELY PLACED BEFORE POURING CONCRETE OR GROUT OR BEFORE APPLYING MORTAR.
4. ALL REINFORCING BARS SHALL BE CLEAN OF RUST, GREASE OR OTHER MATERIAL THAT MAY IMPAIR BOND.

STRUCTURAL STEEL

1. ALL STRUCTURAL STEEL SHALL HAVE A MINIMUM YIELD STRENGTH OF Fy = 248 MPa AND CONFORM WITH ASTM A36 SPECIFICATION.
2. ALL STRUCTURAL STEEL SHALL BE FABRICATED AND ERECTED IN ACCORDANCE WITH THE AISC SPECIFICATIONS AND CODE OF STANDARD PRACTICE.
3. ALL ANCHOR BOLTS, NUTS AND WASHERS, TURNBUCKLES AND TENSION BARS SHALL HAVE A STRENGTH OF ASTM A307.
4. SHOP AND FIELD WELDINGS SHALL BE IN ACCORDANCE WITH AWS D1.1 AND SHALL BE PERFORMED BY QUALIFIED WELDERS.
5. UNLESS OTHERWISE INDICATED, WELDING ELECTRODES SHALL BE E70XX. THE MINIMUM THICKNESS OF WELD SHALL BE 3MM.
6. THE CONTRACTOR SHALL FURNISH ALL PLATES, CLIP ANGLES, SPACERS, CONNECTORS, ETC. REQUIRED FOR THE COMPLETION OF THE WORK EVEN IF SUCH ARE NOT SHOWN IN THE DRAWINGS.

MASONRY

1. ALL MATTERS AND WORKMANSHIP SHALL BE IN ACCORDANCE WITH THE APPLICABLE STANDARDS AND SPECIFICATIONS OF THE NATIONAL CONCRETE MASONRY ASSOCIATION AND UNIFORM BUILDING CODE.
2. CONCRETE MASONRY UNITS SHALL CONFORM TO ASTM C90 GRADE N.
3. UNLESS OTHERWISE INDICATED, CHB IN THIS WORK SHALL HAVE A MINIMUM ULTIMATE COMPRESSIVE STRENGTH AT 28 DAYS OF Fm = 2.7 MPa (400 psi).
4. CONCRETE MASONRY UNITS SHALL BE FILLED SOLIDLY WITH MORTAR. CONCRETE MORTAR AND GROUT FOR ALL REINFORCED MASONRY SHALL CONFORM TO ASTM 270 TYPE M AND SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 10.00 MPa (1500 psi) AT 28 DAYS.
5. PROVIDE A SOLID BAND OF CONCRETE LINTEL BEAM AT THE TOP OF A CHB WALL THAT HAS NO UPPER SUPPORT. THE LINTEL BEAM SHALL HAVE A WIDTH EQUAL TO THE THICKNESS OF THE WALL AND A DEPTH OF NOT LESS THAN 200 MM.
6. ALL MASONRY WALLS SHALL BE REINFORCED ACCORDING TO THE FOLLOWING SCHEDULE OF CONCRETE HOLLOW BLOCK REINFORCEMENT UNLESS OTHERWISE INDICATED IN THE PLANS.

SLAB ON GRADE

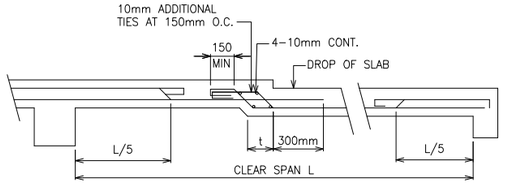
1. THE SOIL SUBGRADE AND FILL LAYERS BELOW ALL SLABS ON GRADE, PAVING AND PIT SLABS SHALL BE MECHANICALLY COMPACTED IN LAYERS TO A MINIMUM OF 95 PERCENT OF THE MODIFIED PROCTOR DENSITY, PER ASTM D1557.
2. ALL SLABS ON GRADE SHALL BE PROVIDED WITH A MINIMUM OF 100 MM THICK COMPACTED CLEAN COARSE SAND BED.
3. UNLESS OTHERWISE NOTED, ALL BEDDED SLABS SHALL BE REINFORCED WITH 10 MM BARS AT 250 MM O.C. EACHWAY AT THE CENTER OF THE SLAB.

COLUMN

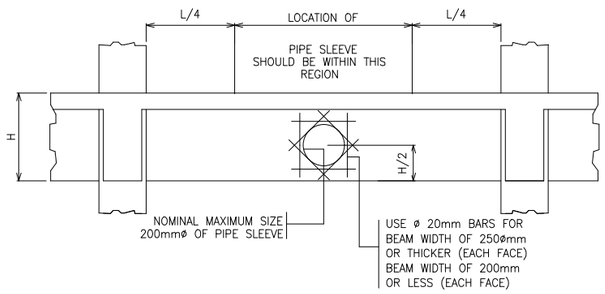
1. WHERE COLUMNS CHANGE IN SIZE, VERTICAL REINFORCEMENTS SHALL BE OFFSET AT A SLOPE NOT MORE THAN 1 ON 6. PROVIDE TRANSVERSE REINFORCEMENT FOR JOINTS WITH BAR OFFSET.
2. LAP SPLICES WHEN REQUIRED, ARE PERMITTED ONLY WITHIN THE CENTER OF THE COLUMN LENGTH AND SHALL BE PROPORTIONED AS TENSION SPLICES. IN NO CASE SHALL THE LAP SPLICE BE LOCATED CLOSER THAN A DISTANCE EQUAL TO THE MAXIMUM COLUMN DIMENSION FROM THE FACE OF THE BEAM-COLUMN JOINT. PROVIDE EXTRA TRANSVERSE REINFORCEMENT OF THE SAME SIZE AND ARRANGEMENT INDICATED IN THE COLUMN SCHEDULE SPACED AT MOST ONE-FOURTH THE MINIMUM COLUMN SECTION DIMENSION THROUGHOUT THE LENGTH OF THE SPLICE OR 100 MM.
3. FOR ALL TIED COLUMNS, PROVIDE TRANSVERSE REINFORCEMENT OF THE SAME SIZE AND ARRANGEMENT INDICATED IN THE COLUMN SECTION SCHEDULE AND SPACED NO GREATER THAN ONE-QUARTER THE MINIMUM COLUMN SECTION DIMENSION NOR 100 MM, OVER A DISTANCE FROM EACH JOINT FACE OF NOT LESS THAN THE LARGER OF THE MAXIMUM COLUMN SECTION DIMENSION, OR ONE-SIXTH OF THE CLEAR HEIGHT OF THE COLUMN OR 450 MM.

BEAM AND GIRDER

1. UNLESS OTHERWISE NOTED IN PLANS OR SPECIFICATIONS, CAMBER ALL BEAMS AT LEAST 8 MM FOR EVERY 450 MM OF SPAN EXCEPT FOR CANTILEVERS FOR WHICH THE CAMBER SHALL BE AS NOTED IN THE PLANS OR AS ORDERED BY THE STRUCTURAL ENGINEER BUT IN NO CASE LESS THAN 19 MM FOR EVERY 3000 MM OF FREE SPAN.
2. IF THERE ARE TWO OR MORE LAYERS OF LONGITUDINAL REINFORCING BARS IN A BEAM OR GIRDER, USE SEPARATORS OF A SIZE NOT LESS THAN 25 MM BARS SPACED ABOUT 900 MM ON CENTER. IN NO CASE SHALL THERE BE LESS THAN TWO (2) SEPARATORS BETWEEN LAYERS OF BARS.
3. LONGITUDINAL REINFORCING BARS SHALL BE PLACED SYMMETRICALLY ABOUT THE VERTICAL CENTER LINE OF THE BEAM OR GIRDER SECTION WHERE POSSIBLE WITH UPPER LAYER BARS PLACED DIRECTLY ABOVE THOSE IN THE BOTTOM LAYER.
4. MINIMUM CONCRETE PROTECTION OF REINFORCING BARS OR SHAPES SHALL BE AS SHOWN IN FIGURE 4. BEAM REINFORCING BARS BOTH TOP AND TERMINATING IN A WALL SHALL EXTEND AT THE MOST 50 MM FROM THE FACE OF THE WALL AND SHALL TERMINATE IN A STANDARD 90°.
5. WHEN A BEAM CROSSES A GIRDER REST BEAM BARS ON TOP OF THE GIRDER BARS. PROVIDE ADDITIONAL TIES WHEN BEAMS OF DIFFERENT TOP ELEVATIONS CROSS A GIRDER.
6. LONGITUDINAL REINFORCEMENTS OF GIRDERS, BOTH TOP AND BOTTOM TERMINATED IN A COLUMN SHALL BE EXTENDED TO THE FAR FACE OF THE CONFINED CONCRETE CORE OF THE COLUMN AND TERMINATED BY A STANDARD 90° HOOK.
7. GENERALLY, NO LAP SPLICE SHALL BE PERMITTED ON BEAMS AND GIRDERS AT POINTS WHERE CRITICAL BENDING STRESSES OCCUR. IN ADDITION, FOR GIRDERS, NO LAP SPLICES SHALL BE LOCATED WITHIN THE JOINTS OR WITHIN A DISTANCE EQUAL TO TWICE THE MEMBER DEPTH FROM THE FACE OF THE JOINT.
8. PROVIDE LAP SPLICES IN GIRDERS WITH HOOP REINFORCEMENT OVER THE LENGTH OF THE LAPPED BARS SPACED NO FARTHER THAN ONE-FOURTH THE NOMINAL DEPTH, OR 100 MM.
9. WELDED SPLICES AND MECHANICAL CONNECTIONS MAY BE USED FOR SPLICING BEAMS AND GIRDERS PROVIDED THAT NO MORE THAN ALTERNATE BARS IN EACH LAYER OF LONGITUDINAL REINFORCEMENT AND SPLICES AT A SECTION AND THE CENTER TO CENTER DISTANCE BETWEEN SPLICES OF ADJACENT BARS IS AT LEAST 600 MM MEASURED ALONG THE AXIS OF THE MEMBER.
10. SEE MECHANICAL, PLUMBING, ELECTRICAL AND FIRE PROTECTION DRAWINGS FOR ALL SUSPENDED AND EMBEDDED PIPING, CONDUITS, DUCTWORKS, EQUIPMENTS, ETC.
11. PIPE AND DUCT SLEEVES SHALL BE LOCATED WITHIN THE REGION BOUNDED BY ONE-FOURTH OF CLEAR SPAN LENGTH FROM THE SUPPORTS.



1 DROP SLAB
NTS



2 TYPICAL PIPE SLEEVE DETAIL
NTS

FOUNDATION

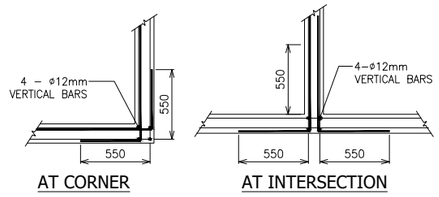
1. FOOTINGS ARE DESIGNED FOR AN ALLOWABLE SOIL BEARING CAPACITY OF 98.17 KPa AT A MINIMUM OF 1.00 METERS FROM THE NATURAL GRADE LEVEL. THE CONTRACTOR SHALL REPORT IN WRITING TO THE STRUCTURAL ENGINEER ON THE ACTUAL SOIL CONDITIONS UNCOVERED AND CONFIRM ACTUAL BEARING CAPACITY OF SOIL BEFORE DEPOSITING CONCRETE.
2. PROVIDE TEMPORARY REMOVAL OF WATER FROM ANY SOURCE DURING CONSTRUCTION. DEWATERING SHALL BE CAREFULLY AND PROPERLY PERFORMED TO AVOID DISTURBING THE FOUNDATIONS AND SLAB BEARING SURFACES.
3. THE CONTRACTOR SHALL DESIGN, INSTALL AND MONITOR EXCAVATIONS RETENTION SYSTEMS AS REQUIRED FOR PROTECTION OF ADJACENT PROPERTIES/STRUCTURES AND PROVIDE ALL MEASURES AND PRECAUTIONS NECESSARY TO MINIMIZE SETTLEMENT AND PREVALENT DAMAGE TO ADJACENT EXISTING OR NEW CONSTRUCTION.
4. PREPARE CONDITIONS OF CONCRETE SUPPLY AND PLACEMENT OF THE COMPLETE FOUNDATION FOR FULL THICKNESS AS A CONTINUOUS MONOLITHIC CASTING.
5. REFER TO ARCHITECTURAL, MECHANICAL, PLUMBING, ELECTRICAL AND OTHER TRADES FOR SUBSOIL DRAINAGE SYSTEM, MACHINERY ANCHORS AND OTHER EMBEDDED ITEMS, DEPRESSIONS, FINISHES, DOWELS FOR MASONRY WALLS, CURBS, ETC.

CONCRETE SLAB

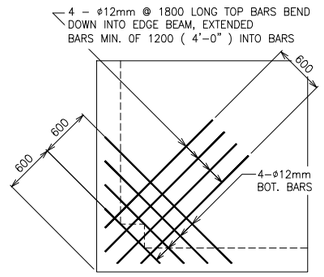
1. ALL SLAB REINFORCEMENTS SHALL HAVE A MINIMUM CLEAR DISTANCE OF 20 MM FROM THE BOTTOM AND FROM THE TOP OF SLABS.
2. UNLESS OTHERWISE DETAILED, FOR CONTINUOUS SLABS WITH THE MAIN REINFORCEMENT RUNNING IN ONE DIRECTION, REINFORCING BARS SHALL BE BENT UP, EXTENDED OR CUT.
3. IF SLABS ARE REINFORCED BOTHWAYS, BARS ALONG THE SHORTER SPAN BE PLACED BELOW THOSE ALONG THE LONGER SPAN AT THE CENTER OF THE SLAB AND BE PLACED OVER THE LONGER SPAN BARS ON AREAS NEAR THE SUPPORTS. THE SPACING OF THE BARS AT THE COLUMN STRIPS SHALL BE APPROXIMATELY ONE AND ONE-HALF (1-1/2) TIMES THAT IN THE MIDDLE STRIPS BUT IN NO CASE GREATER THAN TWO AND ONE-HALF (2-1/2) TIMES THE SLAB THICKNESS OR 450 MM.
4. TEMPERATURE BARS FOR SLABS SHALL BE GENERALLY PLACED NEAR THE FACE IN TENSION AND SHALL NOT BE LESS THAN 0.0025 BT.
5. UNLESS OTHERWISE NOTED, DROP SLABS SHALL BE PROVIDED WITH ADDITIONAL REINFORCEMENT AT THE LOCATION OF DROP.
6. PROVIDE EXTRA REINFORCEMENT FOR CORNER SLAB (TWO ADJACENT DISCONTINUOUS EDGES).
7. SEE MECHANICAL, PLUMBING, ELECTRICAL AND FIRE PROTECTION DRAWINGS FOR ALL SUSPENDED AND EMBEDDED PIPING, CONDUITS, DUCTWORK, EQUIPMENT, ETC.
8. UNLESS OTHERWISE NOTED, EMBEDDED CONDUITS SHALL BE RUN GENERALLY AT MID-BAY AND PARALLEL CONDUITS SHALL BE AT THREE DIAMETERS ON CENTER. CONDUIT SIZE SHALL NOT EXCEED ¼ OF THE SLAB THICKNESS AND SHALL BE LOCATED AT MID THICKNESS OF THE SLAB.

HOLLOW BLOCK REINFORCEMENT SCHEDULE

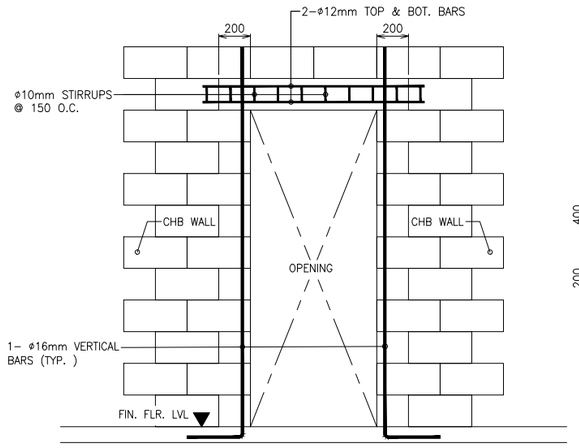
BLOCK THICKNESS (mm)	REINFORCEMENT		NOTES
	HORIZONTAL	VERTICAL	
100	ø10mm @ 600mm O.C.	ø10mm @ 400mm O.C.	1. MINIMUM LAP SPLICE = 30Db 2. PROVIDE 1-180 VERTICAL BAR AT CORNER INTERSECTION END OF WALLS AND EACH SIDE OF OPENING. 3. DOWELS WITH THE SAME SIZE AS VERTICAL OR HORIZONTAL REINFORCEMENT SHALL BE PROVIDED WHERE CHB WALLS ADJOIN COLUMNS, RC BEAMS AND WALL. 4. LINTEL SHALL BEAR AT LEAST 400mm ON EACH SIDE OF MASONRY WALL OPENINGS.
150	ø12mm @ 600mm O.C.	ø12mm @ 400mm O.C.	
200	ø12mm @ 600mm O.C.	ø12mm @ 400mm O.C.	



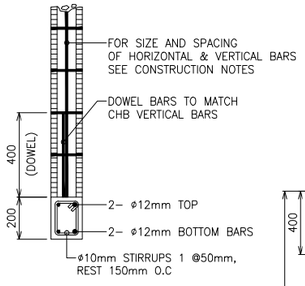
TYP. R.C. WALL CONNECTION DETAIL
SCALE: NTS



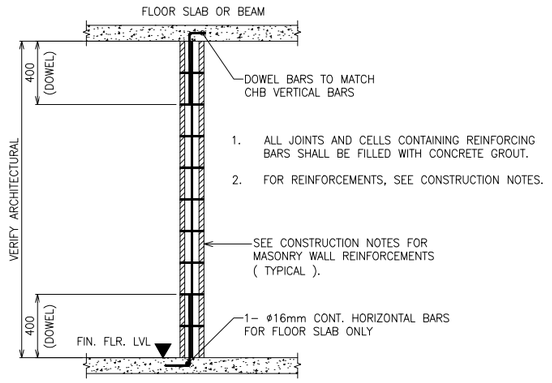
TYP. CORNER SLAB DETAIL
SCALE: NTS



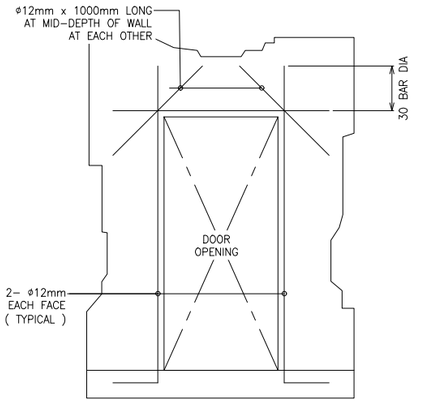
TYP. DETAIL OF LINTEL BEAM AT CHB WALL OPENING
SCALE: NTS



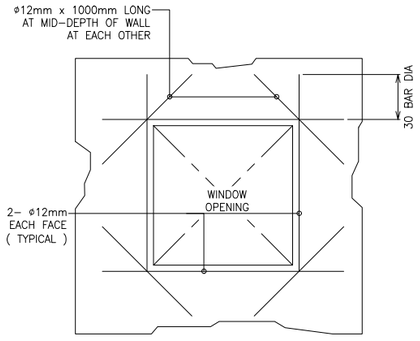
SECTION



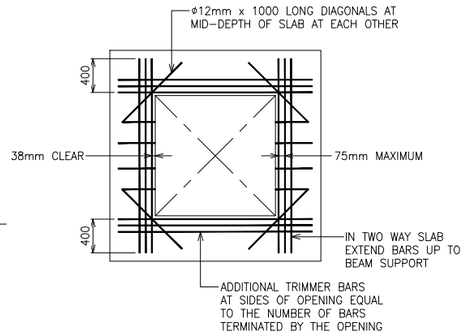
TYP. SECTION OF MASONRY PARTITION REINFORCEMENT
SCALE: NTS



DOOR OPENING DETAIL
SCALE: NTS



WINDOW OPENING DETAIL
SCALE: NTS



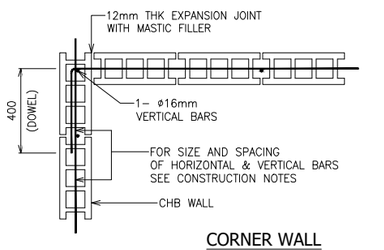
NOTE:
OMIT TRIMMER BARS WHERE OPENING IS FRAMED BY BEAMS.

REINFORCING BAR FOR SLAB OPENING	
OPENING SIZE	VERT. & HOR.
UNDER 600	2 – ⌀12mm
UNDER 800	4 – ⌀12mm

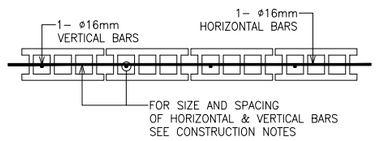
NOTE:

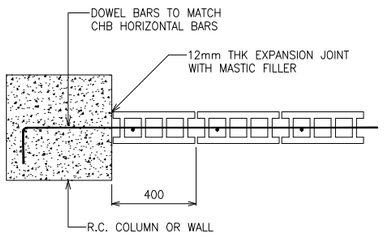
1. PROVIDE THESE ADDITIONAL BARS FOR ALL OPENINGS EXTRA BARS PARALLEL TO THE SIDE OF OPENING EQUAL TO THE NUMBER OF INTERRUPTED BARS BY THE OPENING.
2. SEE ARCHITECTURAL & MECHANICAL PLANS FOR SLAB OPENING LOCATIONS.

SLAB OPENING DETAIL
SCALE: NTS

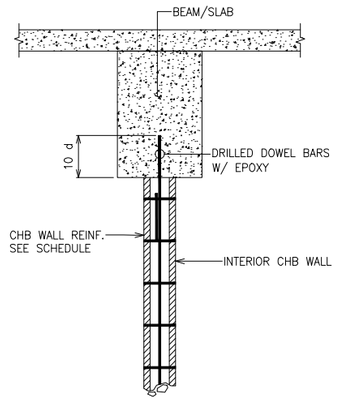


CORNER WALL

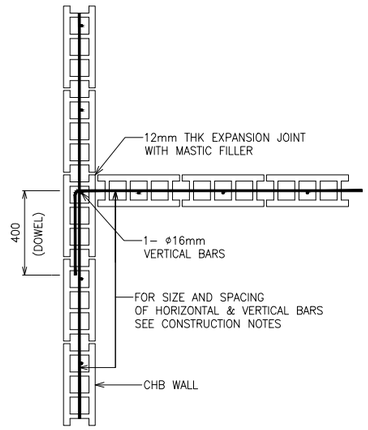




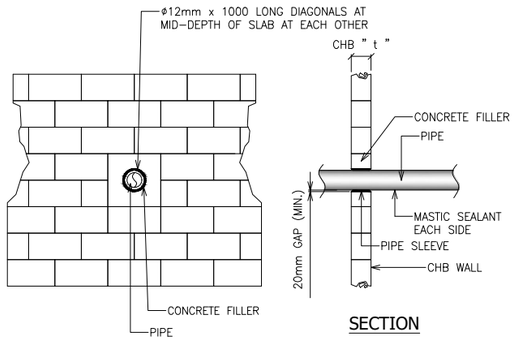
INTERSECTING R.C. COLUMN OR WALL



OPENING OR END WALL



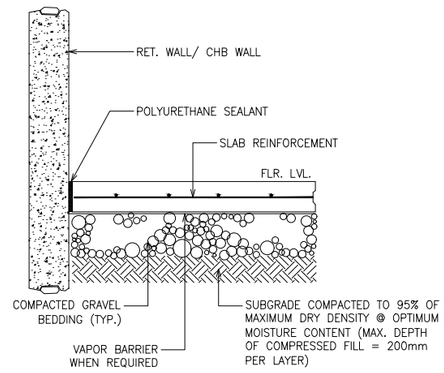
INTERSECTION WALL



SECTION

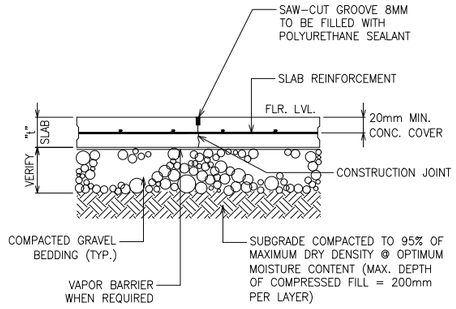
PIPE SLEEVE THRU WALL
SCALE: NTS

TYP. CONNECTION DETAIL OF MASONRY WALL
SCALE : NTS



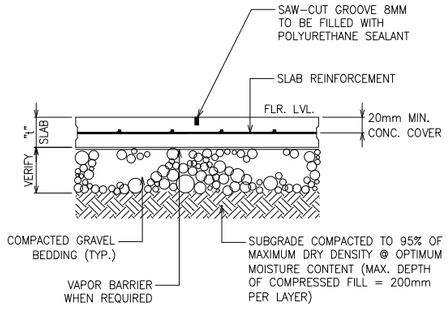
SLAB-ON-GRADE EXPANSION JOINT DETAIL

SCALE: NTS



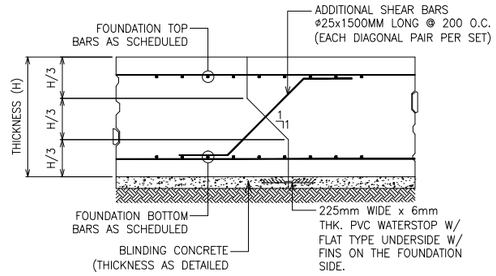
SLAB-ON-GRADE CONSTRUCTION JOINT DETAIL

SCALE: NTS



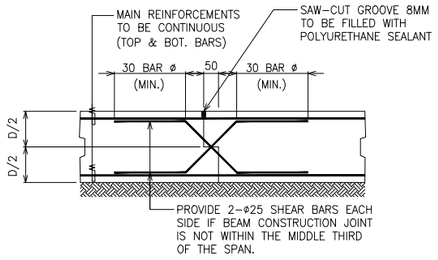
SLAB-ON-GRADE CONTRACTION JOINT DETAIL

SCALE: NTS



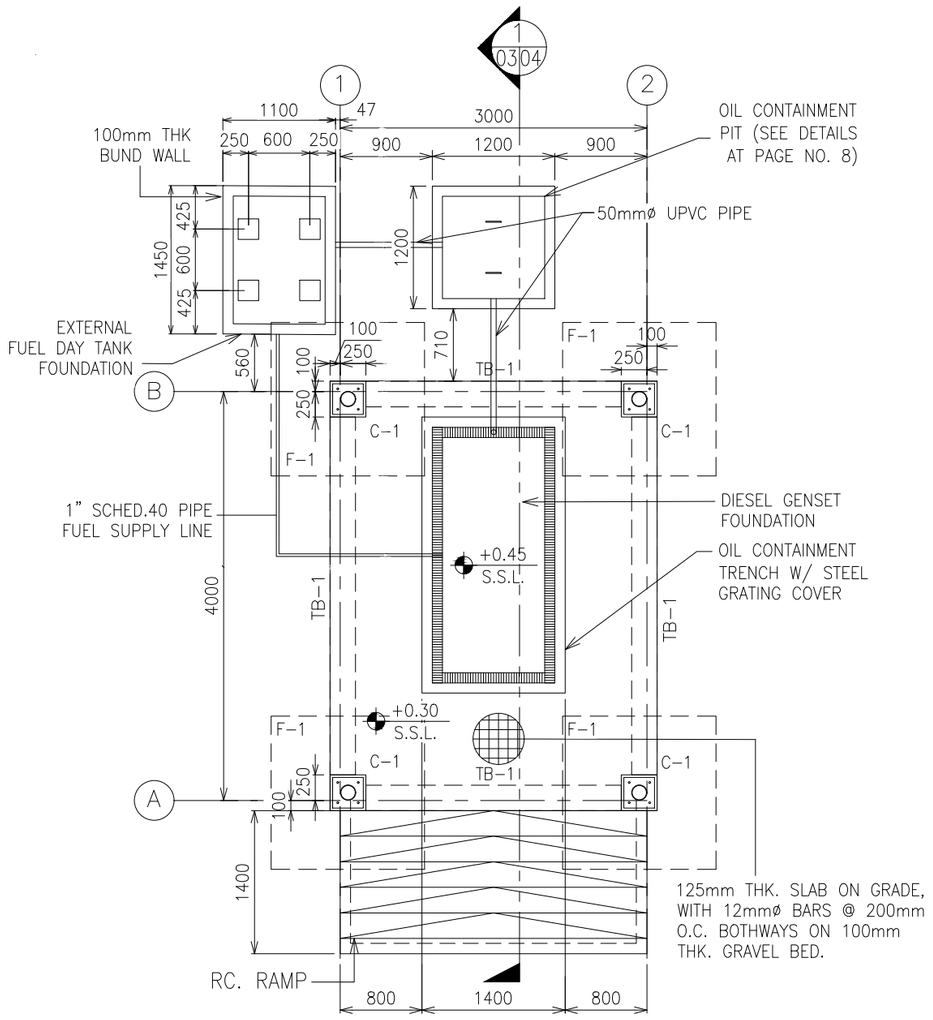
FOOTING CONSTRUCTION JOINT DETAIL

SCALE: NTS



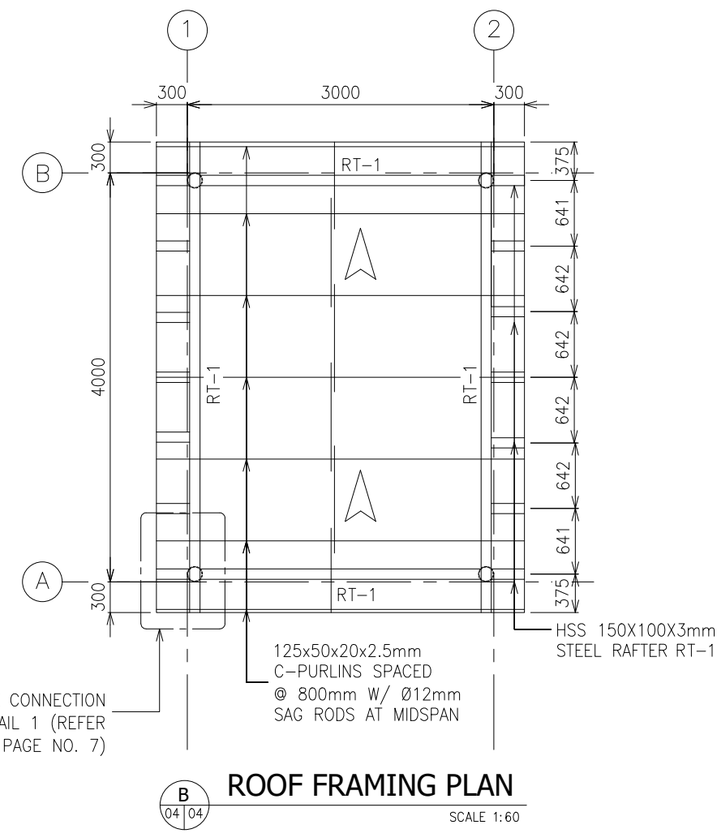
GRADE BEAM CONSTRUCTION JOINT DETAIL

SCALE: NTS



A 04|04 **FOUNDATION PLAN**

SCALE 1:60



B 04|04 **ROOF FRAMING PLAN**

SCALE 1:60

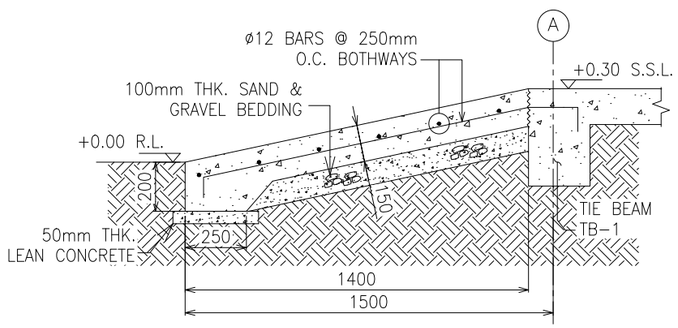
NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. U.O.S., RC PEDESTALS SHALL BE PROVIDED WHERE COLUMNS ARE IN DIRECT CONTACT WITH THE EARTH. THE MINIMUM DIMENSION OF THE PEDESTALS SHALL BE 50mm BIGGER THAN THE COLUMN DIMENSION, MAINTAINING A MINIMUM OFFSET OF 25mm FROM COLUMN FACE.
3. THE DESIGNED NET ALLOWABLE SOIL BEARING CAPACITY USED IS 98.17 kPa AT MINIMUM FOUNDING LEVEL OF 1.00 m BELOW NATURAL GRADE BASED ON BOREHOLE BH–01 OF GEOTECHNICAL REPORT.
4. SITE ADAPTATION: THE GENERAL CONTRACTOR SHALL INFORM THE STRUCTURAL ENGINEER OF ANY SITE CONDITION THAT MAY REQUIRE CHANGES IN THE STRUCTURAL DESIGN.
5. FOR TIE BEAMS REINFORCEMENT REFER TO BEAM SCHEDULE AT SHEET NO. 05
6. FOR FOUNDATION DETAILS REFER TO SHEET NO. 06
7. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.

LEGEND:

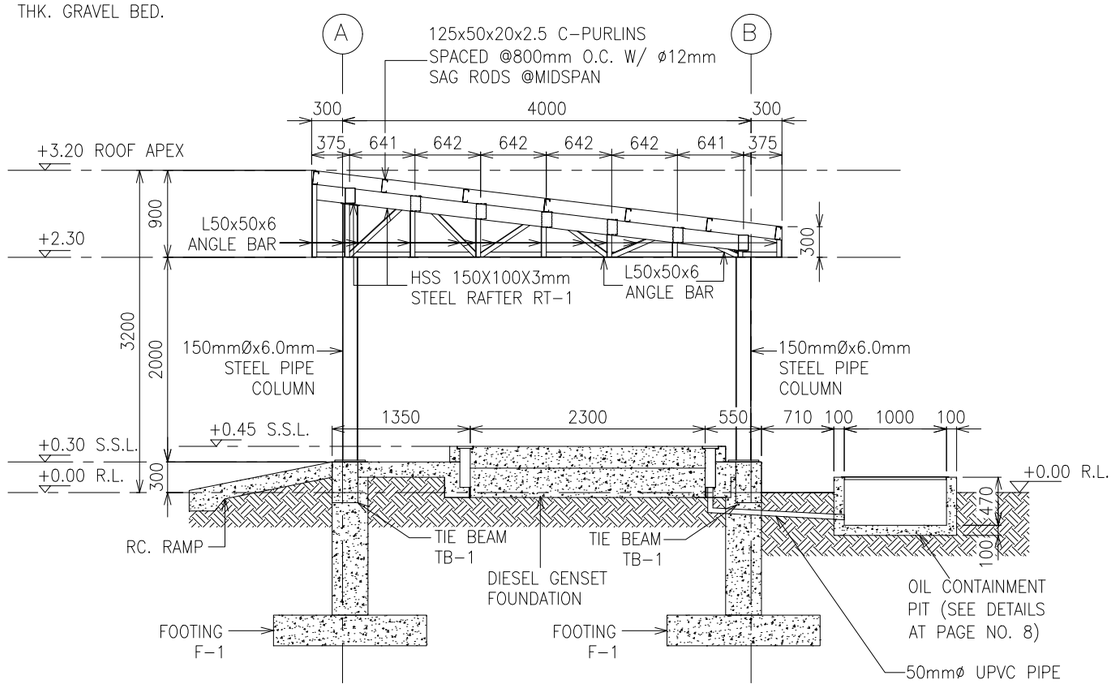
TB–1	TIE BEAM MARK
F1/C1	FOOTING/ COLUMN MARK
RB–1	ROOF BEAM MARK
S.S.L.	STRUCTURAL SLAB LEVEL
F.G.L.	FINISH GRADE LINE
R.L.	ROAD LEVEL

A 01|02 — REFER TO PAGE NO. / PAGE. NO.



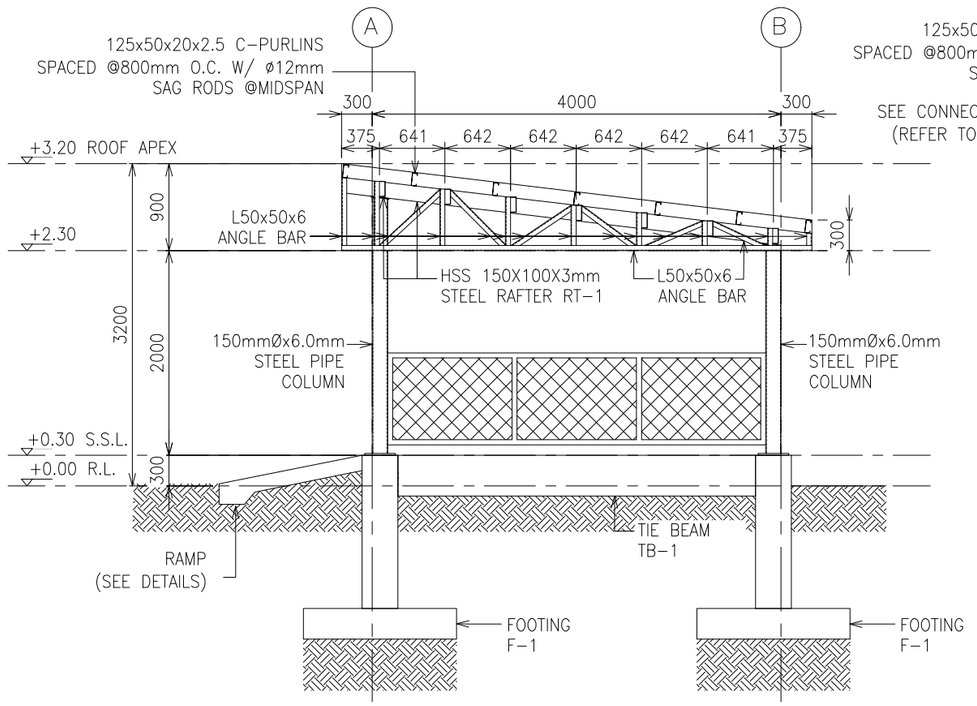
C 04|04 **RAMP DETAIL**

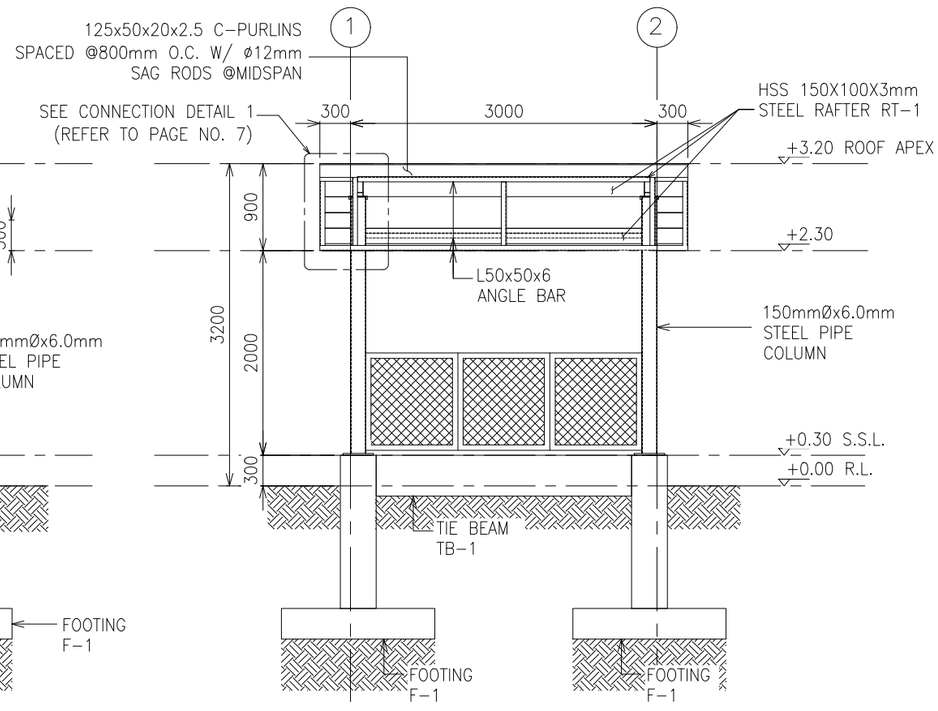
SCALE 1:25



D 04|04 **SECTION 1**

SCALE 1:60





E 05|04 **ELEVATIONS**

SCALE 1:60

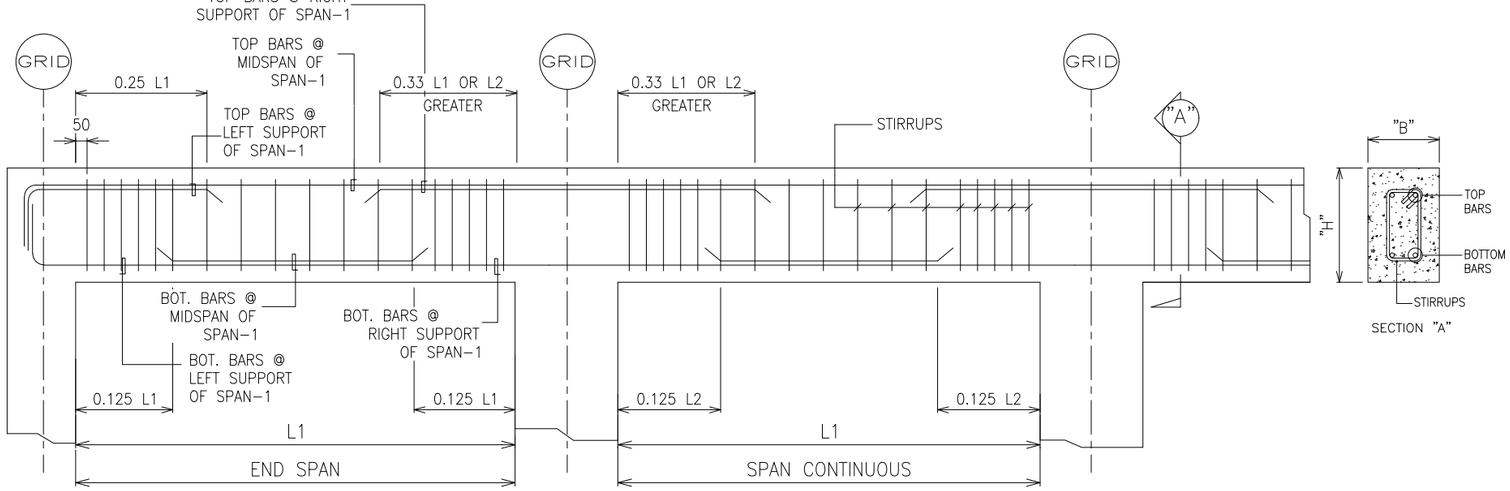
NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. U.O.S., RC PEDESTALS SHALL BE PROVIDED WHERE COLUMNS ARE IN DIRECT CONTACT WITH THE EARTH. THE MINIMUM DIMENSION OF THE PEDESTALS SHALL BE 50mm BIGGER THAN THE COLUMN DIMENSION, MAINTAINING A MINIMUM OFFSET OF 25mm FROM COLUMN FACE.
3. THE DESIGNED NET ALLOWABLE SOIL BEARING CAPACITY USED IS 98.17 kPa AT MINIMUM FOUNDING LEVEL OF 1.00 m BELOW NATURAL GRADE BASED ON BOREHOLE BH-01 OF GEOTECHNICAL REPORT.
4. SITE ADAPTATION: THE GENERAL CONTRACTOR SHALL INFORM THE STRUCTURAL ENGINEER OF ANY SITE CONDITION THAT MAY REQUIRE CHANGES IN THE STRUCTURAL DESIGN.
5. FOR FOUNDATION DETAILS REFER TO SHEET NO. 06
6. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.

LEGEND:

TB–1	TIE BEAM MARK
F1/C1	FOOTING/ COLUMN MARK
RB–1	ROOF BEAM MARK
S.S.L.	STRUCTURAL SLAB LEVEL
F.G.L.	FINISH GRADE LINE
R.L.	ROAD LEVEL

A 01|02 — REFER TO PAGE NO. / PAGE. NO.



TYPE A

SCHEDULE OF STIRRUPS

– 1 SET 10mmø, 1 @ 50, 20 @ 100mm O.C., REST @ 200mm O.C

SCHEDULE OF TIE BEAM

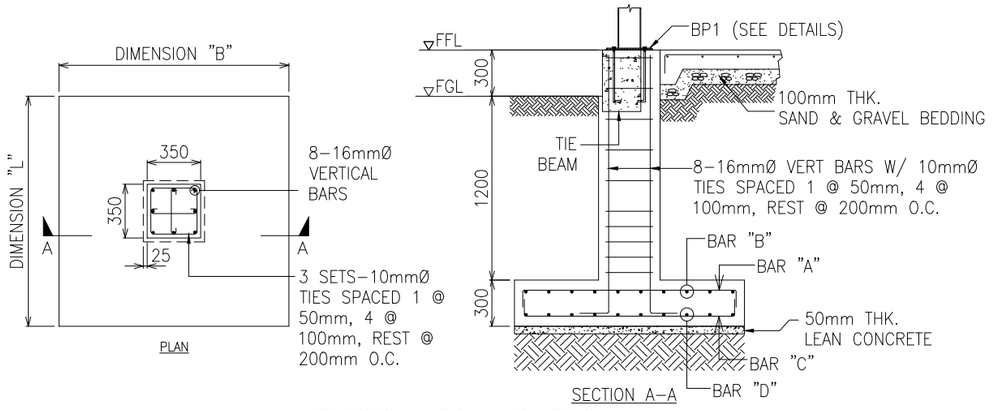
BEAM MARK	SECTION		REINFORCEMENT						WEB BARS (E.F.)	STIRRUPS	REMARKS
			LEFT SUPPORT		MIDSPAN		RIGHT SUPPORT				
	B (mm)	H (mm)	TOP BAR	BOTTOM BAR	TOP BAR	BOTTOM BAR	TOP BAR	BOTTOM BAR			
TB–1	250	400	2–16mmø	2–16mmø	2–16mmø	2–16mmø	2–16mmø	2–16mmø		TYPE–A	

F 05|04 **TYPICAL BEAM DETAILS**

SCALE NTS

FOOTING SCHEDULE

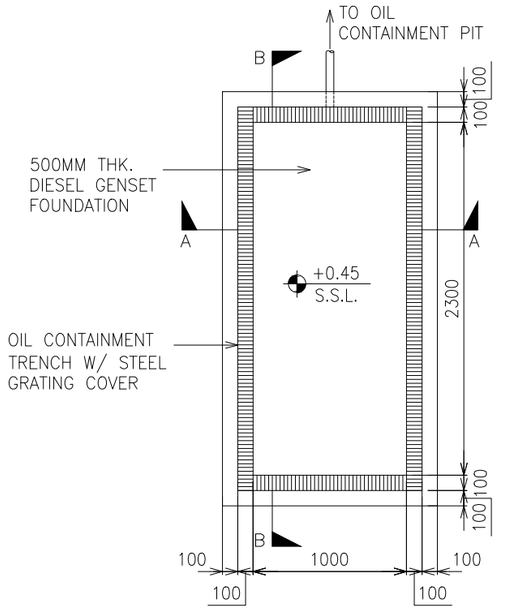
FOOTING MARK	FOOTING DIMENSION (mm)			TOP REINFORCEMENT		BOTTOM REINFORCEMENT	
	DIM. "B"	DIM. "L"	DIM. "T"	BAR "A"	BAR "B"	BAR "C"	BAR "D"
F1	1500	1500	300	10–ø16mm	10–ø16mm	10–ø16mm	10–ø16mm



TYPICAL FOUNDATION DETAILS

G | 06 | 04

SCALE 1:40



DIESEL GENSET FOUNDATION PLAN

J | 06 | 04

SCALE 1:40

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

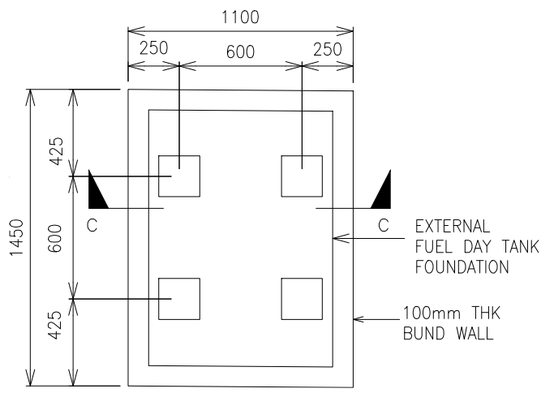
1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. U.O.S., RC PEDESTALS SHALL BE PROVIDED WHERE COLUMNS ARE IN DIRECT CONTACT WITH THE EARTH. THE MINIMUM DIMENSION OF THE PEDESTALS SHALL BE 50mm BIGGER THAN THE COLUMN DIMENSION, MAINTAINING A MINIMUM OFFSET OF 25mm FROM COLUMN FACE.
3. THE DESIGNED NET ALLOWABLE SOIL BEARING CAPACITY USED IS 98.17 kPa AT MINIMUM FOUNDING LEVEL OF 1.00 m BELOW NATURAL GRADE BASED ON BOREHOLE BH–01 OF GEOTECHNICAL REPORT.
4. SITE ADAPTATION: THE GENERAL CONTRACTOR SHALL INFORM THE STRUCTURAL ENGINEER OF ANY SITE CONDITION THAT MAY REQUIRE CHANGES IN THE STRUCTURAL DESIGN.
5. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.
6. ALL CONNECTION ARE WELDED UNLESS OTHERWISE NOTED.
7. MINIMUM THICKNESS OF WELD = 5mm UNLESS OTHERWISE NOTED.
8. U.O.S. ALL STRUCTURAL STEEL MEMBERS SHALL CONFORM TO ASTM SPECIFICATIONS FOR A–36 STEEL.
9. ALL ANCHOR BOLTS, NUTS AND WASHERS, TURNBUCKLES AND TENSION BARS SHALL HAVE A STRENGTH OF ASTM A307.
10. WELDING SHALL CONFORM TO AWS D1.1 STRUCTURAL WELDING CODE STEEL USING E70XX ELECTRODES OR EQUIVALENT.

LEGEND:

TB–1	TIE BEAM MARK
F1/C1	FOOTING/ COLUMN MARK
RB–1	ROOF BEAM MARK
S.S.L.	STRUCTURAL SLAB LEVEL
F.G.L.	FINISH GRADE LINE
R.L.	ROAD LEVEL

A | 01 | 02 ← REFER TO PAGE NO.

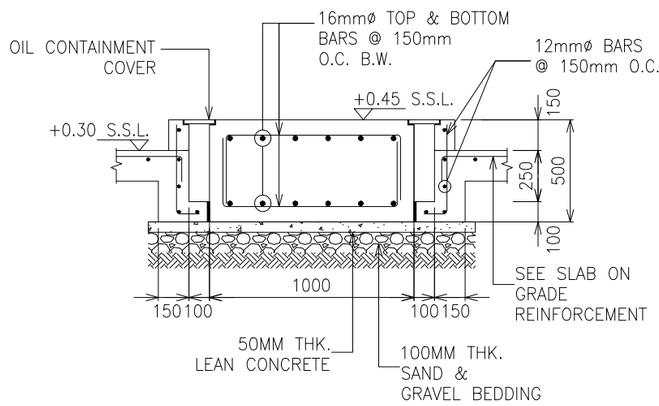
PAGE. NO.



FUEL DAY TANK FOUNDATION PLAN

H | 06 | 04

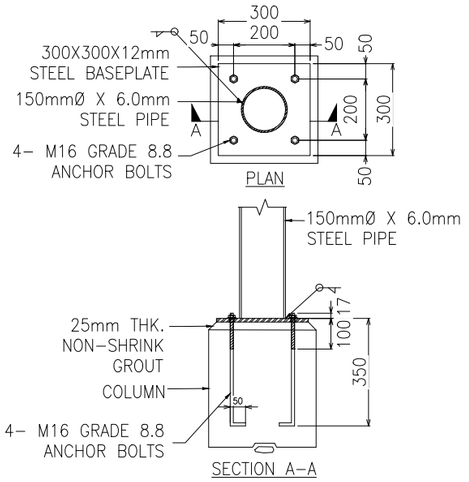
SCALE 1:30



SECTION A-A

K | 06 | 06

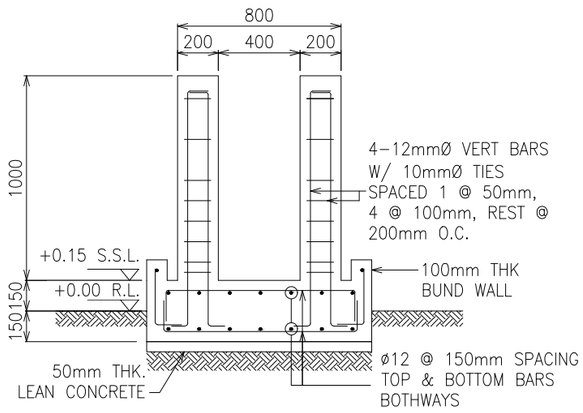
SCALE 1:30



BASEPLATE (BP-1) DETAILS

M | 06 | 06

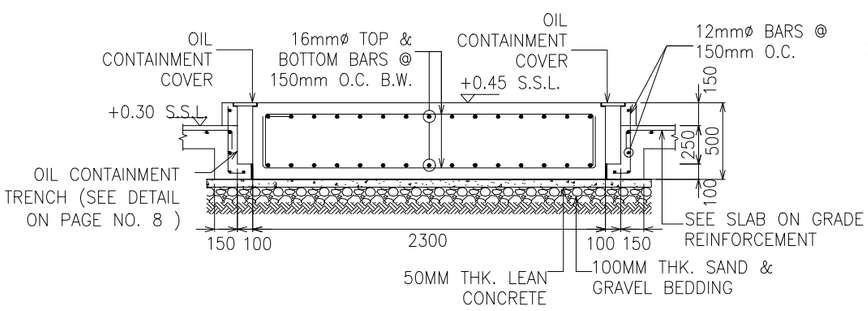
SCALE 1:20



SECTION C-C

I | 06 | 06

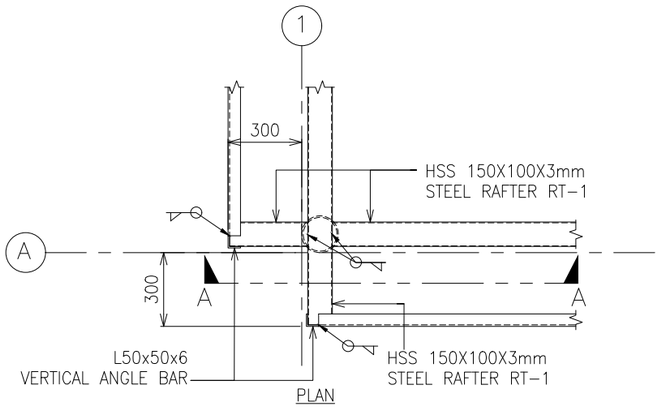
SCALE 1:30



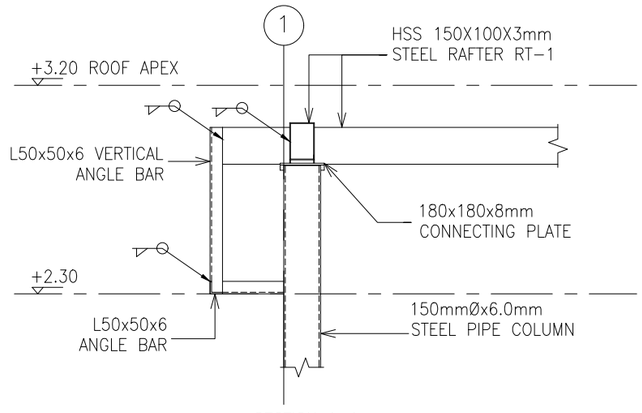
SECTION B-B

L | 06 | 06

SCALE 1:40



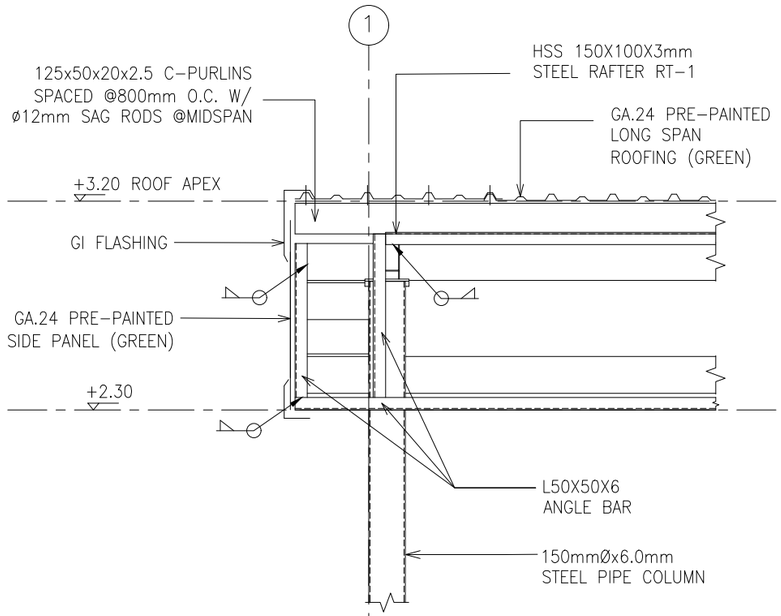
PLAN



SECTION A–A

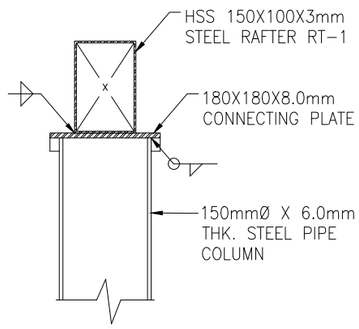
L 07|05 **CONECTION DETAIL 1**

SCALE 1:25



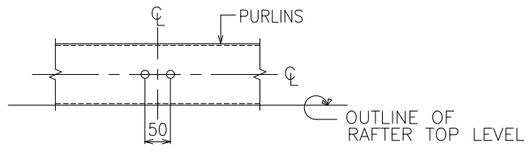
M 07|05 **ROOF CONNECTION DETAILS**

SCALE 1:25



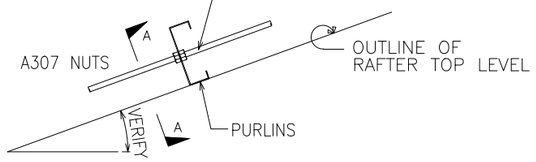
N 07|05 **RAFTER TO COLUMN CONNECTION DETAIL**

SCALE 1:10



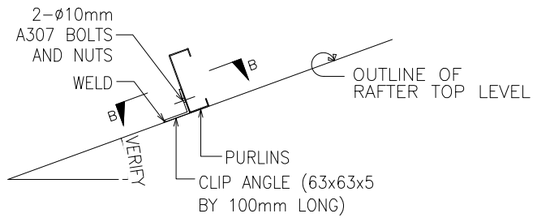
SECTION A–A

12ø SAG RODS FOR LENGTH "L" OF PURLINS
(@ THIRD POINTS FOR L > 2.0m)
(@ MID POINTS FOR L < 2.0m)
(EXTRA @ FREE ENDS OF CANTILEVERS)



O 07|05 **TYP. SAG ROD CONNECTION DETAILS**

SCALE NTS



P 07|05 **PURLIN TO RAFTER CONNECTION DETAILS**

SCALE NTS

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

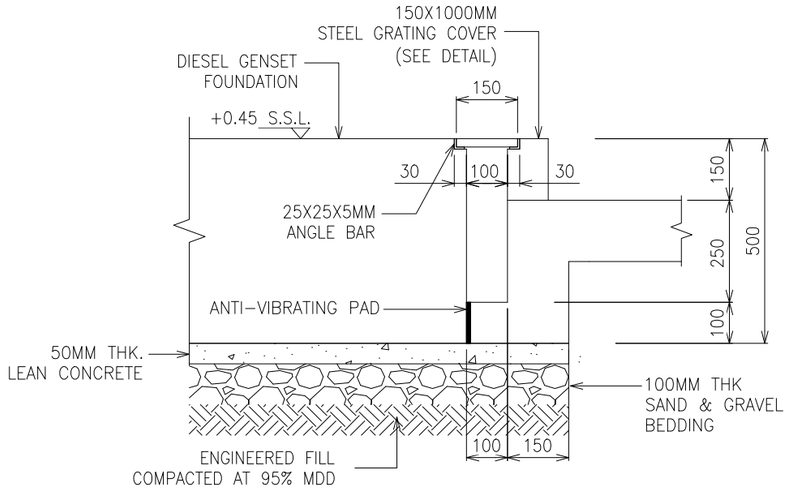
1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. U.O.S., RC PEDESTALS SHALL BE PROVIDED WHERE COLUMNS ARE IN DIRECT CONTACT WITH THE EARTH. THE MINIMUM DIMENSION OF THE PEDESTALS SHALL BE 50mm BIGGER THAN THE COLUMN DIMENSION, MAINTAINING A MINIMUM OFFSET OF 25mm FROM COLUMN FACE.
3. SITE ADAPTATION: THE GENERAL CONTRACTOR SHALL INFORM THE STRUCTURAL ENGINEER OF ANY SITE CONDITION THAT MAY REQUIRE CHANGES IN THE STRUCTURAL DESIGN.
4. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.
6. ALL CONNECTION ARE WELDED UNLESS OTHERWISE NOTED.
7. MINIMUM THICKNESS OF WELD = 5mm UNLESS OTHERWISE NOTED.
8. U.O.S. ALL STRUCTURAL STEEL MEMBERS SHALL CONFORM TO ASTM SPECIFICATIONS FOR A-36 STEEL.
9. ALL ANCHOR BOLTS, NUTS AND WASHERS, TURNBUCKLES AND TENSION BARS SHALL HAVE A STRENGTH OF ASTM A307.
10. WELDING SHALL CONFORM TO AWS D1.1 STRUCTURAL WELDING CODE STEEL USING E70XX ELECTRODES OR EQUIVALENT.

LEGEND:

TB–1	TIE BEAM MARK
F1/C1	FOOTING/ COLUMN MARK
RB–1	ROOF BEAM MARK
S.S.L.	STRUCTURAL SLAB LEVEL
F.G.L.	FINISH GRADE LINE
R.L.	ROAD LEVEL

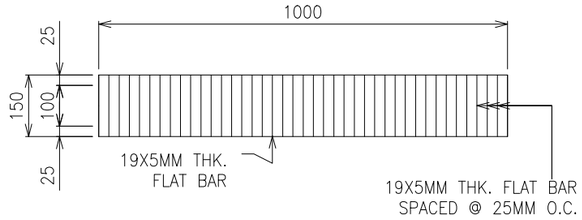
A 01|02 — REFER TO PAGE NO.

PAGE. NO.



Q 08|06 **OIL CONTAINMENT TRENCH DETAIL**

SCALE 1:15



R 08|08 **STEEL GRATING COVER DETAIL**

SCALE 1:15

NOTES: (U.O.S. – UNLESS OTHERWISE STATED)

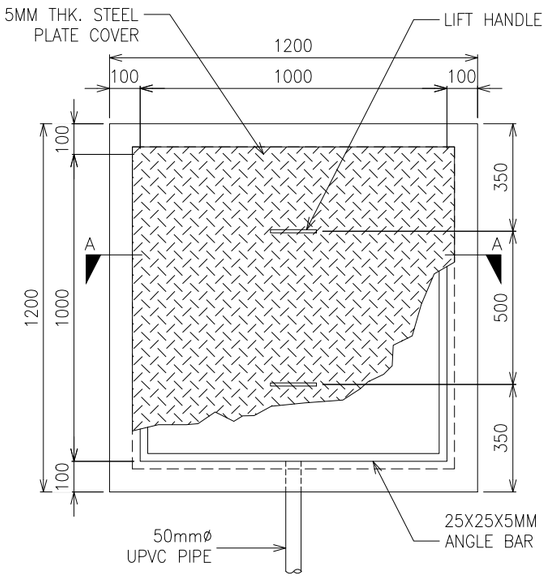
1. ALL DIMENSIONS ARE IN MILLIMETER, U.O.S.
2. U.O.S., RC PEDESTALS SHALL BE PROVIDED WHERE COLUMNS ARE IN DIRECT CONTACT WITH THE EARTH. THE MINIMUM DIMENSION OF THE PEDESTALS SHALL BE 50mm BIGGER THAN THE COLUMN DIMENSION, MAINTAINING A MINIMUM OFFSET OF 25mm FROM COLUMN FACE.
3. SITE ADAPTATION: THE GENERAL CONTRACTOR SHALL INFORM THE STRUCTURAL ENGINEER OF ANY SITE CONDITION THAT MAY REQUIRE CHANGES IN THE STRUCTURAL DESIGN.
4. REFER TO ARCHITECTURAL, MECHANICAL, ELECTRICAL AND OTHER UTILITIES AND EQUIPMENT PLANS FOR THE EXACT SIZE, NUMBER AND LOCATIONS OF ALL SLEEVES OR OPENING THRU FLOOR SLABS, BEAMS AND WALLS AND ALSO BUILDING DIMENSIONS.
6. ALL CONNECTION ARE WELDED UNLESS OTHERWISE NOTED.
7. MINIMUM THICKNESS OF WELD = 5mm UNLESS OTHERWISE NOTED.
8. U.O.S. ALL STRUCTURAL STEEL MEMBERS SHALL CONFORM TO ASTM SPECIFICATIONS FOR A-36 STEEL.
9. ALL ANCHOR BOLTS, NUTS AND WASHERS, TURNBUCKLES AND TENSION BARS SHALL HAVE A STRENGTH OF ASTM A307.
10. WELDING SHALL CONFORM TO AWS D1.1 STRUCTURAL WELDING CODE STEEL USING E70XX ELECTRODES OR EQUIVALENT.

LEGEND:

TB-1	TIE BEAM MARK
F1/C1	FOOTING/ COLUMN MARK
RB-1	ROOF BEAM MARK
S.S.L.	STRUCTURAL SLAB LEVEL
F.G.L.	FINISH GRADE LINE
R.L.	ROAD LEVEL

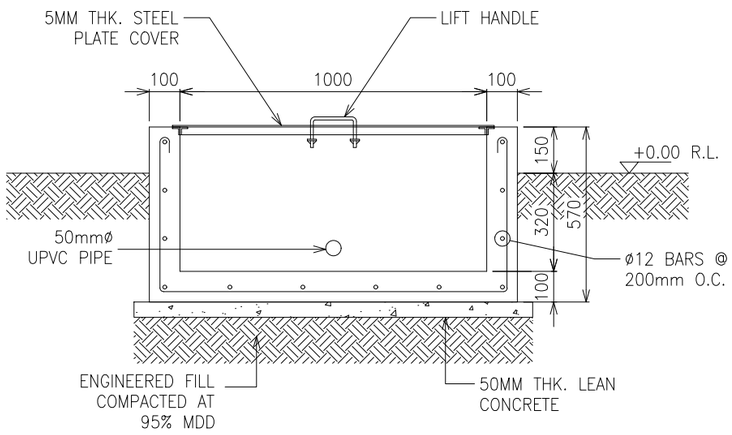
A 01|02 — REFER TO PAGE NO.

PAGE. NO.



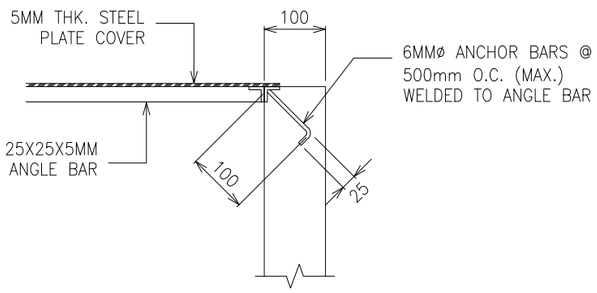
S 08|04 **OIL PIT DETAIL (PLAN VIEW)**

SCALE 1:20



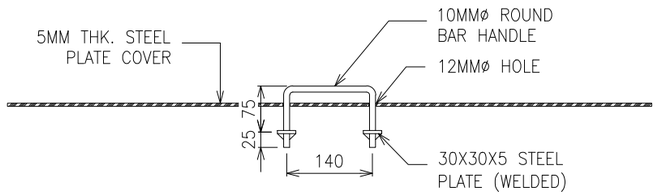
T 08|08 **SECTION A-A**

SCALE 1:20



U 08|08 **PLATE COVER DETAIL**

SCALE 1:10



V 08|08 **LIFT HANDLE DETAIL**

SCALE 1:10